A Virtual Copy of Our BlueROV2: Simulation Results

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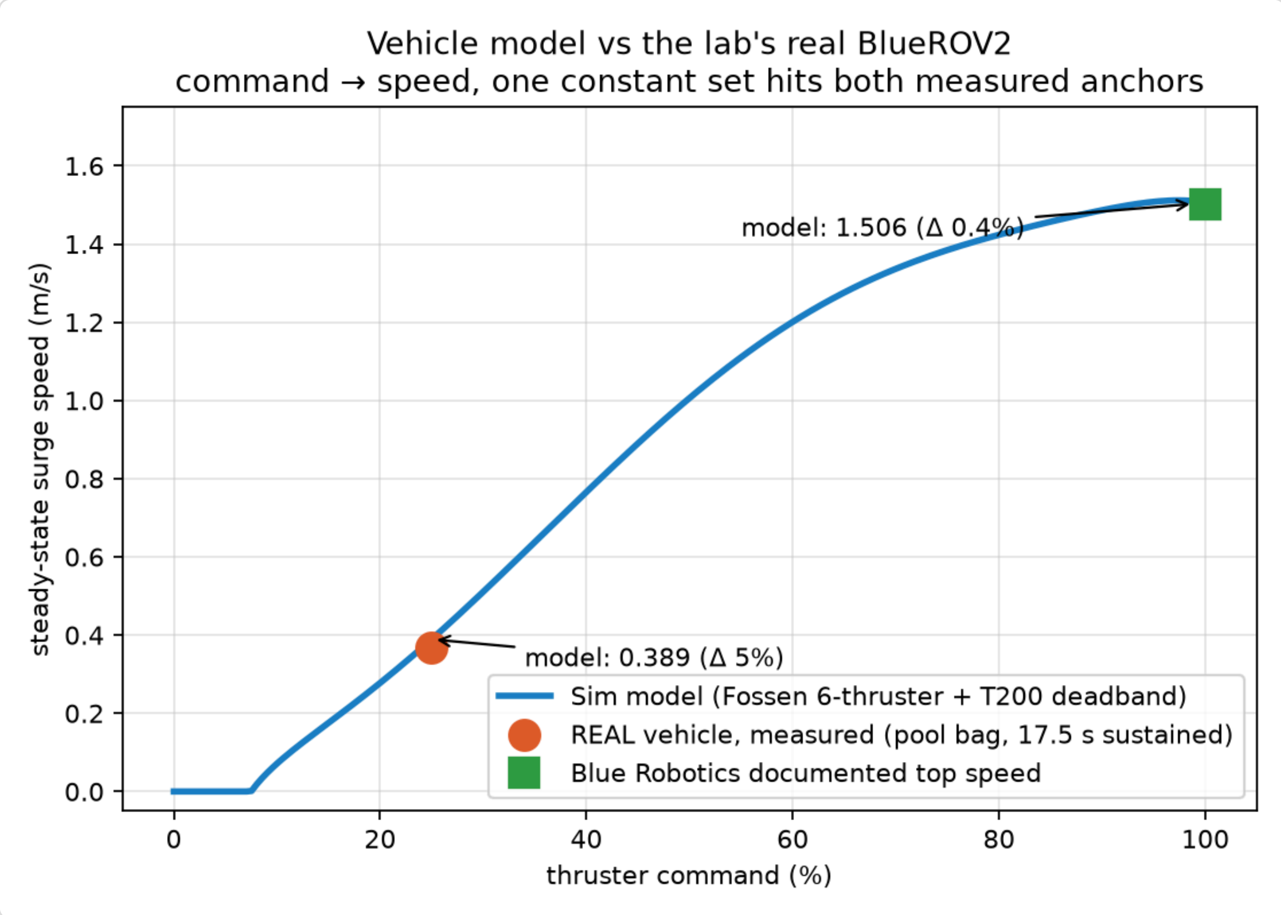
We set up HoloOcean, an underwater robot simulator, on our lab GPU server. We wrote a bridge program that connects it to ROS, the same software framework our real vehicle uses. The simulated robot now publishes sensor data exactly like the real one. Each part was built from real data. For the **body**: our vehicle's flight logs record every thruster, and they confirm it is the standard six-thruster BlueROV2. We built the simulated physics to that exact setup, using Blue Robotics measurements and published studies of this vehicle. For the **movement**: a real pool test recorded every thruster command. We replayed those commands through the simulated physics and compared the result to the speeds the real robot measured. The simulator matches our robot's cruising speed within 5% and its top speed within 1%. For the **sensors**: we read the real recording file and copied every detail into the bridge, including message formats, names, and update rates. Simulated data has the exact same shape as real data. For the **noise**: simulators are unrealistically clean, so we measured the noise of our real sensors from the recording and dialed the same levels into the simulator. Every sensor axis is within 15% of the real vehicle. Finally, to **prove it all works**, we ran our lab's navigation code, completely unchanged, on the simulated robot. It estimated position with about 3% error over distance traveled. That is the same accuracy it achieves on real pool data. We now have a validated virtual copy of our robot for repeatable navigation experiments with perfect ground truth.

1. The simulated vehicle in action

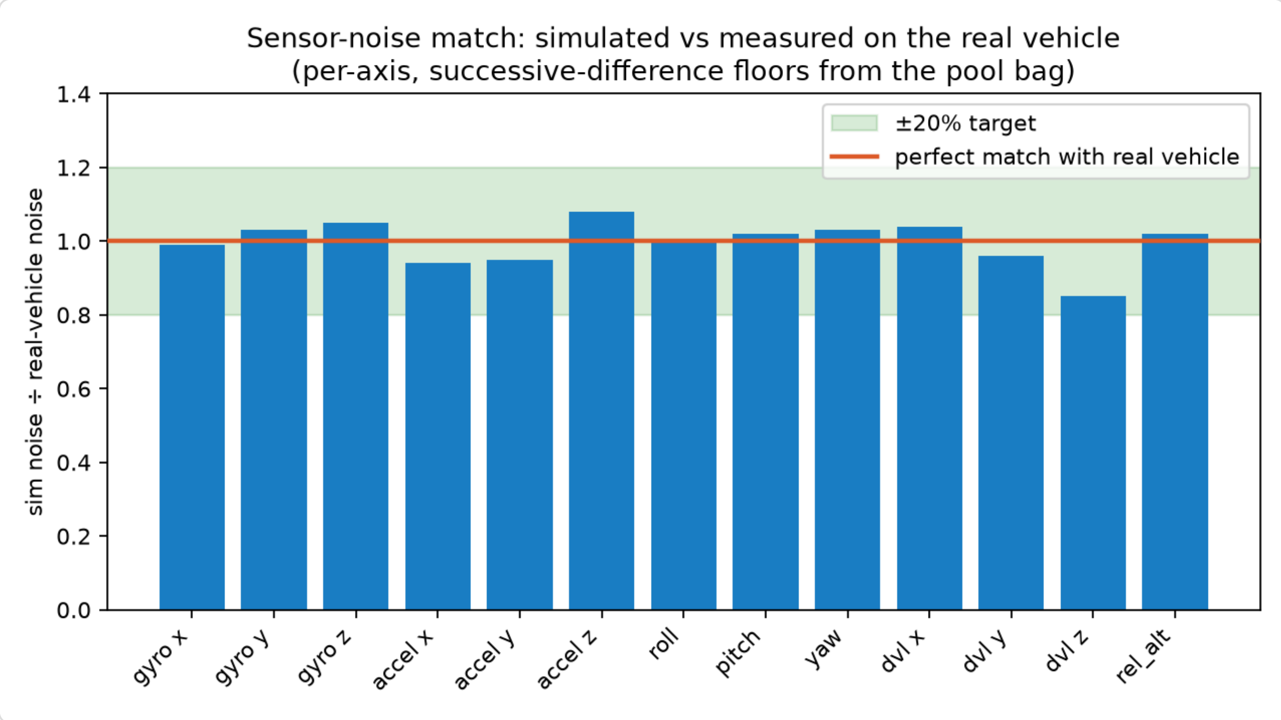
A frame from the simulation video. The BlueROV2 descends, then cruises at about 1 m/s over the seabed while holding depth, driven by the six-thruster physics model. The full 40 second video is provided separately as **simulation_video.mp4**. **Shows: the whole vehicle, physics, and control stack running live.**

2. The simulated robot moves like the real one



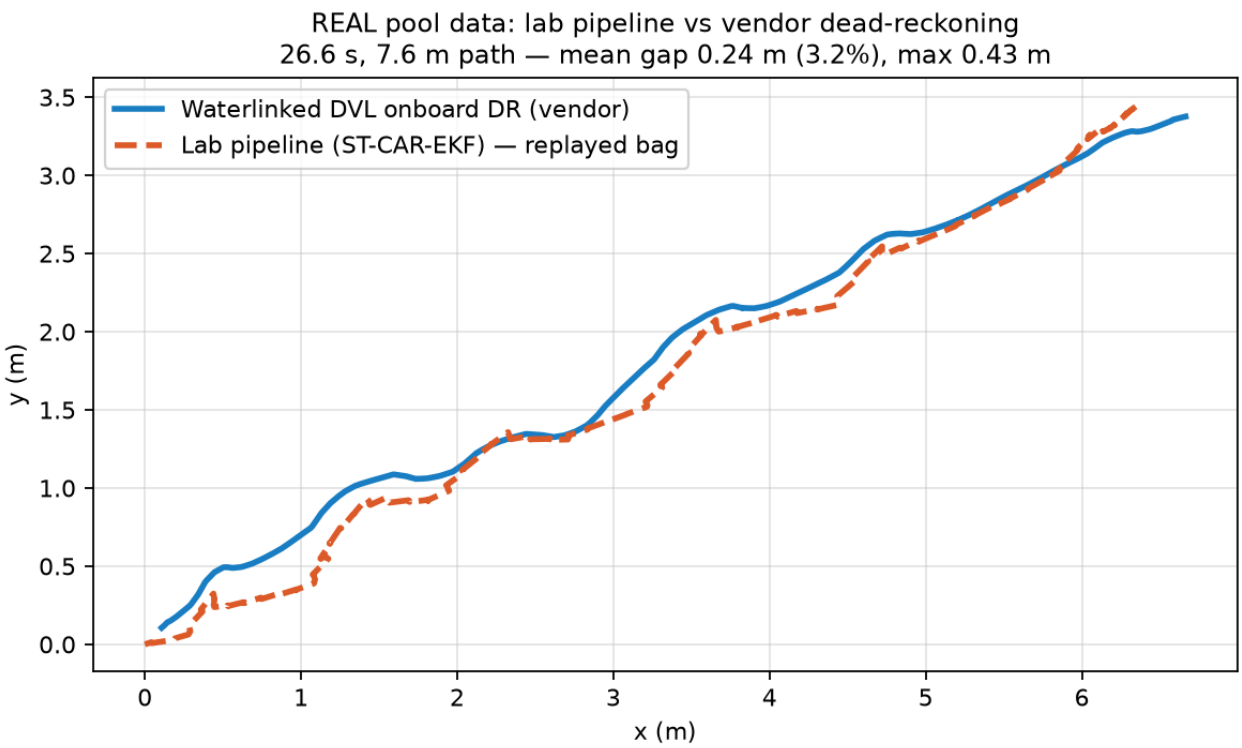
Thruster command vs speed: simulation against the real vehicle. The blue curve is the simulator's predicted speed at each command level. The orange dot is the speed our real robot measured during a sustained 17.5 second segment of the pool test (within 5%). The green square is the manufacturer's documented top speed (within 0.4%). One physics setting hits both points. **Shows: the simulated vehicle behaves like our robot.**

3. The simulated sensors are as noisy as the real ones

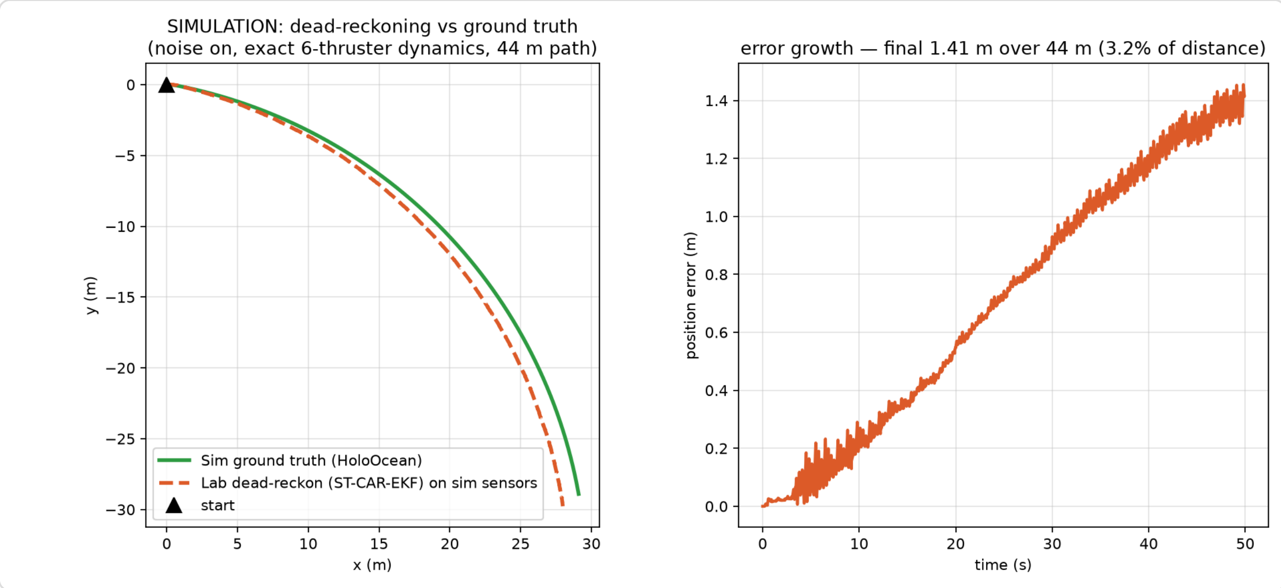


Sensor noise match, axis by axis. Each bar is simulated noise divided by real vehicle noise, so 1.0 means a perfect match. It covers every axis of the IMU, the DVL velocity sensor, and the depth sensor. Real noise was measured from our pool recording using differences between consecutive readings, which stops robot motion from being mistaken for noise. The simulator output was then measured the same way. All 13 axes are inside the 20% band, and most are within 5%. **Shows: the simulator is not too clean. Its data carries our vehicle's real imperfections.**

4. Our navigation code cannot tell the difference



Reference performance on real data. Our pool recording was replayed through the lab's navigation code (orange), shown next to the DVL manufacturer's own onboard position estimate (blue). Two independent systems processed the same swim and agree to a mean gap of 0.24 m over a 7.6 m path, which is 3.2%. **Shows: what correct performance looks like on real data.**



The main result: navigation performance in simulation. Left: the simulator's true path (green) against where our unchanged navigation code believed the robot was (orange), over a 44 m run with realistic noise enabled. Right: position error over time, reaching 1.41 m after 44 m of travel. That is 3.2% of distance, the same error rate the code achieves on real pool data (figure above). **Shows: experiments in this simulator give results that represent the real world, with perfect ground truth available.**